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Model Comparison Report: Logistic Regression vs. SVC

Introduction

This report presents a comparative analysis of two machine learning models—**Logistic Regression with Polynomial Features** and **Support Vector Classifier (SVC)**—built to predict customer purchase behavior.

Data Preparation

An attempt was made to convert the categorical columns Referral and Last_Ad_Seen into numerical form using OneHotEncoder. However, this step was discarded after observing that the model's performance degraded with these encoded features.

On closer inspection, the mean purchase rate within each category of Referral and Last_Ad_Seen was found to lie roughly between **0.25 to 0.30**, indicating that these features lacked strong predictive value. Hence, they were excluded from the final modeling.

The remaining numerical features like Time_on_site, Pages_viewed, Clicked_ad, Cart_value, and Browser_Refresh_Rate were retained. Missing values were not present, and all necessary preprocessing steps like normalization and polynomial feature transformation were applied where needed.

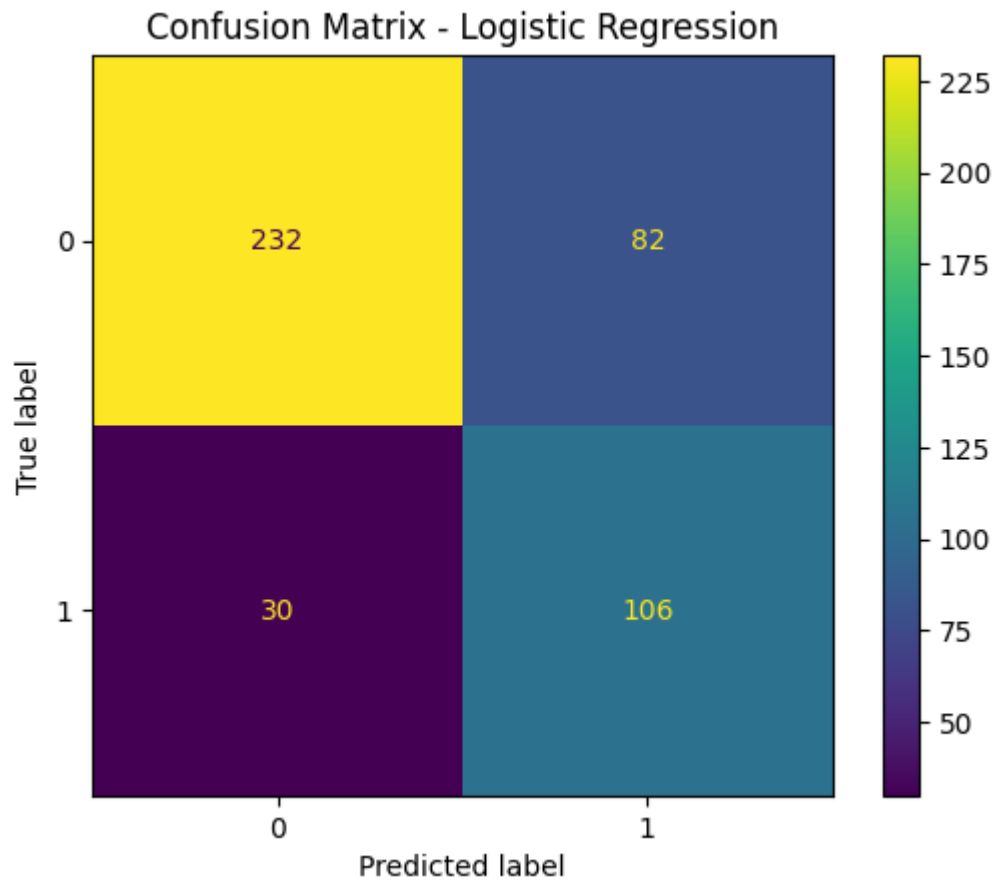
Model 1: Logistic Regression with Polynomial Features

Best Parameters:

- Degree: 2
- Solver: saga
- Penalty: l1
- C: 0.05
- Class Weight: balanced

Performance on Test Data:

- Accuracy: **75.11%**
- F1 Score: **0.65**
- ROC AUC: **0.835**



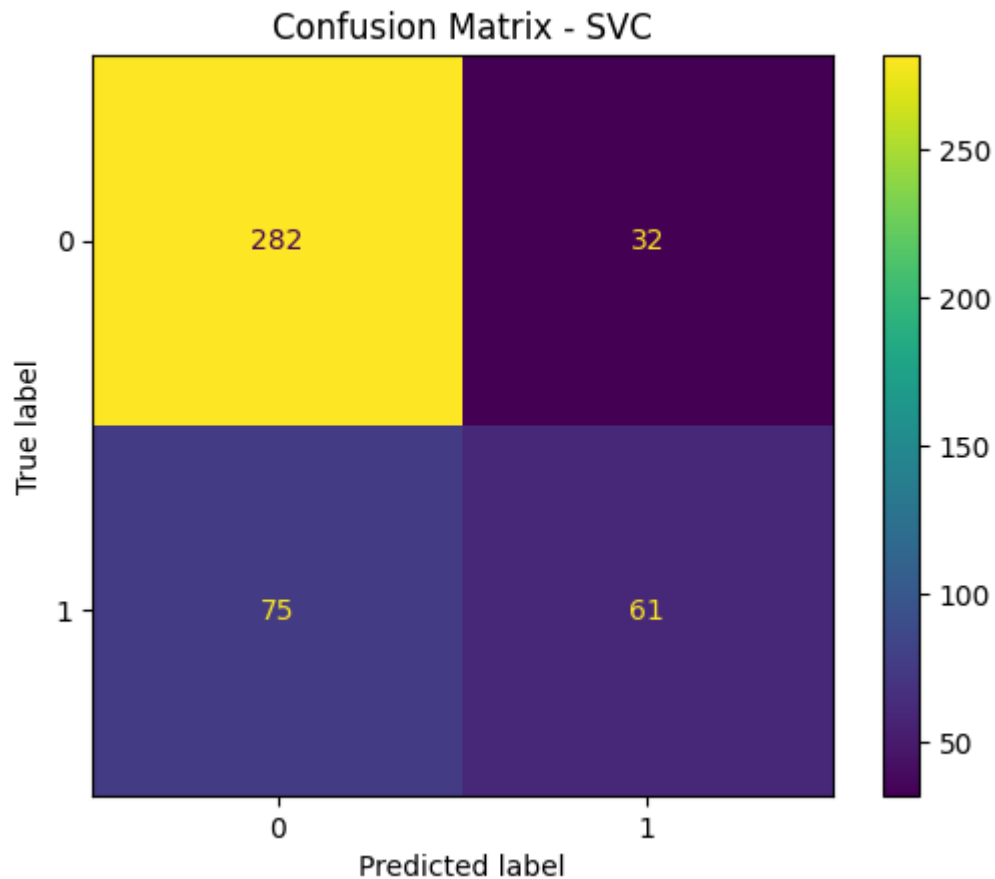
Model 2: Support Vector Classifier (SVC)

Best Parameters:

- Kernel: rbf
- Gamma: 0.1
- C: 1
- Class Weight: None

Performance on Test Data:

- Accuracy: **75.11%**
- F1 Score: **0.65**
- ROC AUC: **0.822**



Conclusion

Both models performed identically in terms of **accuracy** and **F1 score**, but Logistic Regression with Polynomial Features slightly outperformed SVC in terms of **ROC AUC (0.835 vs 0.822)**.

Given its:

- Higher AUC
- Built-in support for interpretability
- Simpler computational structure
- Better handling of class imbalance (with `class_weight='balanced'`)

Logistic Regression with Polynomial Features is the preferred model for this task.